

ELORA RESEARCH STATION, ON, Canada

WMO: 713520

Lat: 43.6420N

Lon: 80.4117W

Elev: 376

StdP: 96.88

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.1	(c) -18.1	(d) -23.9	(e) 0.5	(f) -20.0	(g) -20.7	(h) 0.6	(i) -17.7	(j) 12.5	(k) -4.5	(l) 11.3	(m) -5.7	(n) 2.9	(o) 320	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.0	(c) 29.4	(d) 21.9	(e) 27.8	(f) 21.3	(g) 26.4	(h) 20.4	(i) 23.9	(j) 27.0	(k) 22.6	(l) 26.1	(m) 21.6	(n) 24.9	(o) 4.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.5	25.6	21.6	17.0	24.4	20.5	15.8	23.4	73.8	27.1	68.8	26.3	64.8	25.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.6	8.6	-24.8	31.6	3.5	1.4	-27.3	32.7	-29.3	33.5	-31.3	34.3	-33.8	35.4
			-25.1	25.6	3.4	1.9	-27.6	27.0	-29.6	28.1	-31.5	29.2	-34.0	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-7.1	-6.5	-1.6	5.5	12.2	17.5	19.4	18.6	15.0	8.6	2.1	-3.3
	DBStd	10.61	6.38	5.93	6.15	4.85	4.67	3.69	2.98	2.89	3.93	4.48	4.88	5.20
	HDD10.0	2273	530	463	362	150	29	1	0	0	4	83	238	413
	HDD18.3	4373	788	696	617	386	198	57	23	33	114	303	486	672
	CDD10.0	1096	0	0	3	14	98	227	291	266	155	39	2	0
	CDD18.3	153	0	0	0	0	8	34	56	40	14	1	0	0
	CDH23.3	1328	0	0	1	4	95	303	473	310	133	9	0	0
	CDH26.7	269	0	0	0	0	16	73	102	54	24	1	0	0
Wind	WSAvg	4.0	4.7	4.6	4.5	4.7	4.0	3.4	3.0	2.8	3.1	3.8	4.3	4.6
Precipitation	PrecAvg	889	74	58	56	77	83	83	87	74	77	79	78	69
	PrecMax	1136	123	121	136	130	157	182	154	162	153	138	152	125
	PrecMin	647	38	28	18	27	24	25	19	17	21	22	30	32
	PrecStd	133	25	24	30	31	30	38	39	33	34	33	30	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	10.0	19.7	23.6	28.5	30.7	31.1	30.6	29.4	24.4	16.3	11.6
		MCWB	9.7	7.8	15.2	16.5	19.5	22.9	23.5	22.5	21.5	17.9	12.9	10.5
	2%	DB	6.1	5.6	13.8	19.6	26.1	28.9	29.3	28.1	26.7	21.1	13.9	7.9
		MCWB	5.5	4.4	10.2	13.8	18.6	21.5	22.1	21.3	20.1	16.2	12.0	6.8
	5%	DB	3.3	3.1	10.1	16.8	23.5	26.9	27.6	26.5	24.6	18.4	11.8	5.4
		MCWB	2.6	2.0	7.9	11.9	17.0	21.2	21.5	20.7	19.1	15.1	9.9	4.5
	10%	DB	1.3	1.2	6.8	13.8	20.9	24.8	26.0	25.0	22.4	16.0	9.5	3.0
		MCWB	0.7	0.4	4.8	9.6	15.2	19.6	20.3	19.8	17.9	13.4	7.9	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	8.5	15.7	18.1	21.6	26.3	26.2	24.7	23.0	19.7	14.2	10.5
		MCDB	9.8	9.6	19.6	22.0	26.4	28.5	28.8	27.4	27.1	22.4	15.0	11.6
	2%	WB	5.6	4.5	11.1	15.0	19.7	23.3	24.1	23.0	21.4	17.3	12.3	7.2
		MCDB	5.9	5.4	12.9	18.4	24.4	26.7	27.0	26.3	25.3	19.5	13.7	7.7
	5%	WB	2.7	2.1	7.9	12.8	18.1	21.9	22.7	21.8	20.1	15.4	10.0	4.5
		MCDB	3.2	2.9	10.0	15.5	22.4	25.6	26.0	25.0	23.0	18.1	11.6	5.3
	10%	WB	0.7	0.5	4.9	10.2	16.3	20.4	21.5	20.8	18.8	13.7	8.0	2.3
		MCDB	1.2	1.2	6.7	13.2	19.6	23.5	24.6	23.6	21.6	15.8	9.5	2.9
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	8.6	10.0	11.0	10.5	11.0	11.0	11.3	9.3	7.1	6.1
		MCDBR	8.0	9.2	12.7	15.1	14.3	13.0	12.7	12.7	13.9	13.1	10.9	7.9
		MCWBR	7.8	8.2	9.7	10.3	8.2	7.5	6.9	7.0	8.0	8.6	8.8	7.3
	5% WB	MCDBR	7.8	8.7	12.4	13.4	12.8	11.7	10.9	10.8	11.6	11.8	10.0	7.8
		MCWBR	7.7	8.1	10.0	10.8	8.2	8.0	7.2	7.0	7.4	8.4	8.6	7.5
Clear-Sky Solar Irradiance	taub		0.283	0.326	0.322	0.357	0.382	0.412	0.404	0.394	0.363	0.347	0.304	0.283
	taud		2.361	2.191	2.389	2.342	2.316	2.247	2.288	2.305	2.387	2.420	2.499	2.425
	Ebn at Noon		848	854	919	907	889	860	862	860	862	829	823	816
	Edh at Noon		78	111	104	119	126	135	128	122	104	88	69	68
All-Sky Solar Radiation	RadAvg		1.38	2.24	3.51	4.45	5.38	5.93	5.91	5.14	4.11	2.45	1.50	1.00
	RadStd		0.18	0.28	0.36	0.49	0.52	0.40	0.39	0.26	0.39	0.24	0.21	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	N/A	-1.20	-1.52	-124	-179	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page